

## Tests & Quizzes

### Quiz 7

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#### Part 1 of 1 4.0 / 5.0 Points

Question 1 of 5

1.0

1.0 Points

Suppose the current block height is 915,600 and a transaction has `nLockTime` = 917,616. Additionally, a separate UTXO has a CLTV lock of `L` = 917,616. What is the precise behavioral difference between these two locks?

- ✓ A. `nLockTime` prevents the transaction from being broadcast to the network until block 917,616, whereas CLTV prevents the UTXO from being created until block 917,616
- ✓ B. `nLockTime` prevents the transaction itself from being mined into any block before height 917,616, whereas CLTV prevents a child transaction from spending that output until height 917,616
- ✓ C. `nLockTime` locks the output for a relative delay of 917,616 blocks after confirmation, whereas CLTV applies an absolute lock at the transaction level
- ✓ D. Both `nLockTime` and CLTV prevent the transaction from being mined, but `nLockTime` applies at the output level while CLTV applies at the transaction level

**Answer Key:** B

Question 2 of 5

1.0

1.0 Points

In the HTLC payment routing scenario, Bob creates secret  $S$  and computes  $H = \text{SHA256}(S)$ . Alice creates an HTLC locked to  $H$  in her channel with Charlie, and Charlie creates an HTLC locked to the same  $H$  in his channel with Bob. Bob spends Charlie's HTLC output by revealing  $S$ . What happens next and why is the overall scheme trustless?

- ✓ A. Once Bob reveals  $S$  on-chain to spend Charlie's output,  $S$  becomes publicly visible, so Charlie learns  $S$  and can spend Alice's HTLC output
- ✓ B. Charlie must now ask Bob for  $S$  through a separate message before he can spend Alice's HTLC, which introduces a trust dependency on Bob's cooperation
- ✓ C. Alice automatically refunds Charlie using her CSV time lock path as soon as Bob reveals  $S$ , eliminating Charlie's need to act
- ✓ D.  $S$  remains hidden even after Bob spends Charlie's output because witness data in SegWit transactions is not publicly visible on-chain

**Answer Key:** A

Question 3 of 5

0.0

1.0 Points

A parent transaction confirmed at block 1000 has a CSV lock of 100 blocks on its output. A child transaction is created that spends this output and sets  $nSequence = 100$ . At which block height can this child transaction first be successfully mined?

- ✓ A. Block 1000, because  $nSequence$  alone without CSV is sufficient to enforce the relative delay from the moment the parent is confirmed
- ✗ B. Block 1100, because the output has aged exactly 100 blocks since its confirmation at block 1000, satisfying the relative lock condition
- ✓ C. Block 1101, because the output must reach an age of at least 100 blocks meaning it needs 100 confirmations after block 1000, making it spendable starting at block 1101
- ✓ D. Block 1001, because  $nSequence = 100$  signals the miner to include it after just one confirmation of the parent transaction

**Answer Key:** C

Question 4 of 5

1.0

1.0 Points

Non-collaborative transaction batching is not possible in Bitcoin but is possible in Ethereum through rollups. What is the precise technical reason this limitation exists in Bitcoin?

- ✓ A. Bitcoin's block size limit is too small to accommodate merged transactions, making batching physically impossible
- ✓ B. The UTXO model prevents combining independently signed transactions because any attempt to merge them would invalidate their signatures
- ✓ C. Bitcoin nodes reject any transaction in the mempool that references inputs already claimed by another pending transaction, making batching a double-spend by definition
- ✓ D. Bitcoin's scripting language is non-Turing complete, which prevents the execution of the batching logic required to combine independently signed transactions

**Answer Key:** B

Question 5 of 5

1.0

1.0 Points

In the naïve revocation key construction for commitment transaction C1a held by Alice, Bob generates a key pair ( $r_b$ ,  $R_b$ ) and uses  $R_b$  as the revocation key in C1a. What is the fundamental flaw in this approach?

- ✓ A. Alice can derive  $r_b$  from  $R_b$  using elliptic curve mathematics, allowing her to spend her own output immediately without waiting for the time lock
- ✓ B. The revocation key  $R_b$  is visible on-chain, allowing any third party to steal Alice's funds by computing the private key from the public key
- ✓ C. Bob already knows  $r_b$ , so he can claim Alice's time-locked output even when she honestly publishes the latest commitment
- ✓ D. Bob cannot verify whether the revocation key has been used because  $R_b$  is a public key visible to all network participants

**Answer Key:** C